

AI-BASED SMART ENERGY MONITORING SYSTEM

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by

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SIKKIM MANIPAL INSTITUTE OF TECHNOLOGY
MAJITAR, EAST SIKKIM - 737136, JUNE 2026**

Certificate

This is to certify that the project report entitled “AI-BASED SMART ENERGY MONITORING SYSTEM ” submitted by ANKIT KUMAR (Reg No. 202200158) to Sikkim Manipal Institute Of Technology, Sikkim in partial fulfillment for the award of degree of Bachelor of Technology in Electronics And Communication Engineering, is a bonafide record of the project work carried out by him under my guidance and supervision during the Academic Session January– May, 2026.

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ABSTRACT

Rapid growth in the use of electrical appliances has significantly increased electricity demand in residential environments. Conventional household energy meters generally report only total electricity usage and provide little information about the contribution of individual devices. This lack of appliance-level visibility makes it difficult for consumers to identify high-energy equipment, recognize abnormal consumption patterns, and make informed decisions regarding energy conservation.

The work presented in this report introduces an intelligent residential energy analytics platform based on Non-Intrusive Load Monitoring (NILM). Using the publicly available UK-DALE dataset, a Bidirectional Long Short-Term Memory (BiLSTM) network is trained to learn temporal characteristics from combined household power signals. The processed time-series data is transformed into sequential input windows, enabling the model to estimate the electricity demand of selected appliances, including refrigerators, washing machines, dishwashers, televisions, and kettles [3], [4].

Beyond device-level energy estimation, the framework incorporates abnormal consumption identification, appliance performance analysis, and electricity cost calculation. A Streamlit-based interface provides visual access to usage trends, predicted appliance demand, anomaly notifications, and estimated electricity expenses, allowing users to better understand and monitor household energy behaviour.

Experimental results indicate that the BiLSTM network is capable of extracting meaningful appliance-level information from aggregate power measurements without requiring dedicated sensors for each device. By integrating deep learning, NILM, and interactive analytics, the developed framework offers a practical approach for enhancing energy awareness and supporting more efficient electricity consumption decisions [1], [2], [5], [6].

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Abbreviations

LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
NILM	Non-Intrusive Load Monitoring
ILM	Intrusive Load Monitoring
BiLSTM	Bidirectional Long Short-Term Memory
LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
DNN	Deep Neural Network
ML	Machine Learning
DL	Deep Learning
UK-DALE	UK Domestic Appliance-Level Electricity Dataset
MAE	Mean Absolute Error
MSE	Mean Squared Error
RMSE	Root Mean Square Error
F1 Score	Harmonic Mean of Precision and Recall
IoT	Internet of Things
GUI	Graphical User Interface
API	Application Programming Interface
CSV	Comma-Separated Values
CPU	Central Processing Unit
GPU	Graphics Processing Unit
RAM	Random Access Memory
kWh	Kilowatt-Hour
UI	User Interface
KPI	Key Performance Indicator
SQL	Structured Query Language
ReLU	Rectified Linear Unit
HDF5	Hierarchical Data Format Version 5
Keras	Deep Learning API for TensorFlow
ETA	Estimated Time of Arrival
UTC	Coordinated Universal Time
SMIT	Sikkim Manipal Institute of Technology

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Chapter 1

INTRODUCTION

1.1 Background & Motivation

The increasing dependence on electrical appliances has transformed the way energy is consumed in modern households. Devices such as refrigerators, televisions, washing machines, kitchen appliances, and entertainment systems operate for extended periods and collectively contribute to a significant portion of residential electricity demand. As electricity usage continues to grow, consumers are becoming more interested in understanding where and how energy is consumed within their homes.

Access to detailed consumption information plays an important role in promoting efficient electricity utilization. Knowledge of device-wise usage can assist households in identifying inefficient appliances, reducing unnecessary consumption, and controlling monthly electricity expenses. However, obtaining such information using conventional monitoring approaches often requires additional hardware installations, increasing both implementation cost and maintenance requirements.

Recent developments in data analytics and artificial intelligence have enabled new approaches for examining residential electricity behaviour. Instead of relying solely on dedicated sensing infrastructure, computational techniques can be employed to analyze household consumption records and derive meaningful information regarding appliance operation. These approaches provide an opportunity to obtain detailed consumption insights using existing electricity measurements while minimizing deployment complexity [1], [2].

At the same time, advances in deep learning have significantly improved the ability of intelligent systems to recognize patterns within sequential data. Modern neural network architectures have demonstrated strong performance in applications involving forecasting, classification, and behavioural analysis of time-series information [3], [4], [5], [6]. Their capability to learn complex temporal relationships makes them particularly suitable for residential electricity analysis.

Motivated by these developments, this study investigates a data-driven approach for extracting device-level consumption information from household electricity records. The developed framework combines deep learning, analytical processing, and interactive visualization to provide detailed insight into residential electricity usage and support more informed energy-related decision-making.

1.2 Need for the Project

The increasing complexity of household electricity consumption has created a growing demand for more detailed and accessible energy information. Although residential consumers receive periodic electricity bills and overall consumption statistics, they often lack sufficient knowledge about how different appliances contribute to total electricity usage. This limitation makes it difficult to identify inefficient consumption practices and prioritize corrective actions.

Many households experience avoidable electricity expenditure due to prolonged appliance operation, unnoticed energy wastage, and changing usage habits. Without access to detailed consumption insights, users may be unable to determine which devices are responsible for increased electricity demand or unexpected variations in monthly energy costs. Consequently, opportunities for improving efficiency and reducing expenditure are frequently overlooked.

Existing appliance-monitoring solutions often rely on additional hardware installations and dedicated measurement devices. While such approaches can provide detailed information, their deployment may involve higher costs, increased maintenance requirements, and additional implementation complexity. These factors can limit their practicality for large-scale residential adoption.

At the same time, growing interest in sustainable energy utilization has highlighted the importance of tools capable of converting raw electricity data into meaningful information. Consumers increasingly require solutions that not only provide consumption breakdowns but also assist in identifying unusual usage behaviour, understanding long-term trends, and evaluating the financial impact of electricity consumption.

Therefore, there is a need for an integrated analytical framework capable of transforming household electricity records into actionable insights. Such a solution should provide detailed consumption visibility, support informed decision-making, and assist users in improving electricity utilization without imposing significant infrastructure requirements.

1.3 Problem Statement

- Conventional electricity metering systems provide only overall household consumption information and do not indicate the contribution of individual appliances.
- Consumers often lack sufficient information to identify devices responsible for excessive electricity usage and increased energy expenditure.
- Unusual operating behaviour and inefficient appliance usage may remain unnoticed due to the absence of detailed consumption visibility.
- Hardware-based monitoring approaches require additional sensing equipment, resulting in higher installation costs and increased maintenance requirements.
- Many available solutions focus primarily on consumption measurement and provide limited support for comprehensive analysis and decision-making.
- Users generally have limited access to detailed consumption insights that can assist in understanding usage behaviour and improving electricity utilization.
- The absence of an integrated monitoring and analytical platform restricts the ability to effectively evaluate household electricity consumption patterns.

1.4 Objectives

- To design a deep learning-based prediction framework for estimating the electricity usage of individual household appliances from combined consumption measurements.
- To utilize a publicly available residential electricity dataset for model development, testing, and performance assessment.
- To examine device-wise consumption behaviour and generate detailed usage insights from household electricity records.
- To identify unusual consumption events through automated analysis techniques.
- To estimate electricity expenditure based on projected appliance usage and configurable tariff rates.
- To develop an interactive visualization environment for presenting analytical results, consumption statistics, and monitoring information.
- To improve user understanding of residential electricity usage through data-driven analysis and visual reporting.
- To evaluate the effectiveness of the developed framework using quantitative performance measures and comparative analysis.

1.5 Scope of the Project

This study is limited to the analysis of residential electricity consumption using household energy records obtained from a publicly available dataset. The developed framework focuses on deriving device-level consumption information from combined household electricity measurements and examining the behaviour of selected household appliances.

The work encompasses consumption estimation, usage analysis, abnormal event identification, expenditure assessment, and visualization of analytical results. The developed framework is evaluated using historical electricity records and is intended to demonstrate the feasibility of applying data-driven techniques for residential electricity analysis.

The current implementation considers a limited set of household appliances, namely Refrigerator, Washing Machine, Dishwasher, Television, and Kettle. The evaluation is performed using data obtained from the UK-DALE dataset, and the generated results are applicable within the context of the selected dataset and experimental setup.

Real-time hardware integration, smart meter deployment, Internet of Things (IoT) connectivity, and edge-device implementation are beyond the scope of the present work. These aspects, along with support for additional appliances and larger datasets, are considered potential areas for future enhancement.

The developed framework provides a foundation for residential electricity analysis and can be extended to support broader monitoring and energy-management applications in future studies.

1.6 Organization of the Report

This report is structured into seven chapters, each addressing a specific aspect of the study.

Chapter 1 introduces the research topic and presents the background, project need, problem statement, objectives, scope, and overall organization of the report.

Chapter 2 reviews relevant research related to residential electricity analysis, energy disaggregation approaches, deep learning applications, and intelligent monitoring solutions.

Chapter 3 discusses the analysis and design aspects of the developed framework, including system architecture, workflow, and major functional components.

Chapter 4 describes the methodology adopted for data preparation, feature extraction, sequence construction, model development, training procedures, and performance assessment.

Chapter 5 presents the implementation of the developed framework, covering software tools, processing modules, analytical components, and visualization features.

Chapter 6 discusses the experimental findings obtained during evaluation and includes performance analysis, prediction results, anomaly analysis, expenditure assessment, and visualization outputs.

Chapter 7 summarizes the major conclusions of the study and outlines potential directions for future research and enhancement.

Chapter 2

LITERATURE SURVEY

2.1 Introduction

Research related to residential electricity analysis has expanded significantly in recent years due to increasing interest in energy efficiency, intelligent monitoring, and data-driven decision-making. A variety of approaches have been proposed to extract meaningful information from household electricity data, ranging from traditional statistical methods to advanced artificial intelligence techniques [1], [2]. These developments have contributed to the creation of systems capable of providing detailed insights into electricity usage behaviour and appliance operation.

Earlier studies primarily relied on signal-processing techniques, rule-based approaches, and manually engineered features to interpret electricity consumption patterns. Although these methods demonstrated promising results in controlled environments, their effectiveness often decreased when applied to real-world household data characterized by complex usage behaviour and overlapping appliance activity [2], [3].

The emergence of machine learning and deep learning has introduced more sophisticated solutions for electricity consumption analysis. Modern learning frameworks are capable of automatically extracting relevant features from large datasets and identifying complex relationships within time-series data. As a result, significant improvements have been reported in prediction accuracy, pattern recognition, and device-level consumption estimation [4], [5].

In parallel, researchers have explored complementary functionalities such as abnormal usage identification, expenditure analysis, consumption forecasting, and visualization techniques. These developments have broadened the scope of residential electricity analytics beyond basic consumption estimation and have increased the practical value of intelligent monitoring systems [6].

This chapter presents a review of relevant studies related to residential electricity analysis, energy disaggregation techniques, deep learning applications, and intelligent monitoring frameworks. The literature survey provides an understanding of existing approaches, highlights their strengths and limitations, and identifies research opportunities that motivate the development of the framework presented in this study.

2.2 Literature Survey

Sl. No.	Author	Title of the Paper	Journal / Conference / Book	Findings	Gap Analysis	Relevance to the Project
1	Huzzat et al. (2025)	A Survey of Traditional and Emerging Deep Learning Techniques for NILM	AI Journal (MDPI)	Presents a detailed overview of NILM research and discusses the evolution of deep learning techniques such as CNNs, LSTMs, and Transformers for appliance-level energy estimation.	The work mainly reviews existing approaches and does not implement a complete monitoring framework for practical deployment.	Provided background knowledge on modern NILM methodologies and guided the selection of deep learning techniques for the proposed system.
2	Huber et al. (2021)	Review on Deep Neural Networks Applied to Low-Frequency NILM	Energies (MDPI)	Examines the application of deep neural network models for low-frequency NILM data and evaluates their effectiveness in energy disaggregation tasks.	The study concentrates on model evaluation and lacks features related to visualization, analytics, and end-user interaction.	Helped establish the suitability of deep learning models for appliance-level energy prediction from aggregate power measurements.
3	Kelly and Knottenbelt (2015)	Neural NILM: Deep Neural Networks Applied to Energy Disaggregation	ACM e-Energy Conference	Introduced deep learning-based NILM approaches and demonstrated the potential of neural networks for improving energy disaggregation performance.	The research primarily emphasizes model development and provides limited support for user-oriented monitoring applications.	Served as one of the foundational studies influencing the adoption of deep learning methods in NILM systems.

Sl. No.	Author	Title of the Paper	Journal / Conference / Book	Findings	Gap Analysis	Relevance to the Project
4	Zhou et al. (2021)	Non-Intrusive Load Monitoring Using a CNN-LSTM-RF Model Considering Label Correlation and Class-Imbalance	IEEE Access	Proposed a hybrid framework combining CNN, LSTM, and Random Forest models to enhance appliance classification and energy estimation accuracy.	The hybrid architecture increases computational complexity and offers limited integration with practical energy management tools.	Demonstrated the effectiveness of sequence-based learning methods for improving appliance recognition and disaggregation accuracy.
5.	Zai et al. (2023)	Non-Intrusive Load Monitoring Based on the Combination of Gate-Transformer and CNN	Electronics (MDPI)	Utilized Transformer and CNN architectures to capture complex consumption patterns and improve NILM prediction performance.	The approach requires significant computational resources and focuses mainly on prediction accuracy rather than complete energy monitoring solutions.	Provided insights into recent advancements in NILM architectures and highlighted the importance of intelligent energy monitoring systems.
6.	Zhang et al. (2018)	Sequence-to-Point Learning with Neural Networks for NILM	AAAI Workshop / NILM Research	Introduced a sequence-to-point learning framework capable of estimating appliance-level consumption directly from aggregate electricity signals.	The study mainly targets energy disaggregation and does not address analytics, anomaly detection, or visualization requirements.	Supported the adoption of temporal learning techniques for appliance-level energy estimation in the proposed system.

Table 1. Literature Survey Table

2.3 Comparative Analysis of Existing Works

Ref no.	Technique used	Strengths	Limitations
[1]	Deep Learning-Based NILM Survey	Provides a broad understanding of conventional and modern NILM methodologies and research trends.	Does not provide validation through an integrated monitoring platform
[2]	Deep Neural Networks for NILM	Demonstrates the effectiveness of DNN architectures for appliance-level energy disaggregation.	Limited emphasis on practical residential deployment
[3]	CNN and LSTM-Based NILM Models	Shows the potential of deep learning techniques for extracting appliance consumption patterns.	Primarily targets prediction accuracy rather than end-user analytics
[4]	CNN-LSTM-RF Hybrid Approach	Effectively captures complex temporal relationships and improves prediction accuracy.	Higher computational requirements may limit practical deployment
[5]	Transformer-CNN Architecture	Captures long-range dependencies and improves prediction accuracy	Emphasizes prediction performance with limited consideration of user-oriented features
[6]	Sequence-to-Point Learning	Provides direct appliance-level estimation from aggregate power measurements with good accuracy.	Lacks integrated analytical, visualization, and decision-support capabilities

Table 2. Comparative Analysis of the reviewed literature

2.4 Research Gap

The literature review demonstrates that substantial progress has been achieved in residential electricity analysis through the application of advanced learning techniques and energy disaggregation methods. Existing studies have reported improvements in device-level consumption estimation and appliance identification; however, several limitations remain from a practical implementation perspective.

A review of the selected studies reveals that most research efforts concentrate primarily on prediction performance and benchmarking. Comparatively less attention has been given to the development of comprehensive frameworks that combine analytical, monitoring, and decision-support functionalities within a single environment.

In addition, features such as abnormal consumption analysis, expenditure assessment, appliance-wise usage reporting, and user-oriented visualization are not consistently integrated into many existing solutions. As a result, the practical usability of several proposed approaches remains limited for everyday residential applications.

Another observed limitation is the lack of interactive platforms that enable users to interpret analytical results and monitor electricity usage in an accessible manner. In many cases, outputs are presented mainly through performance metrics and experimental evaluations rather than through tools designed to support end-user engagement.

Based on these observations, there exists an opportunity to develop a unified residential electricity analytics framework that combines device-level consumption estimation, anomaly analysis, expenditure calculation, and interactive visualization. Addressing these gaps can improve the practical value of electricity monitoring systems and provide users with more actionable consumption insights.

Chapter 3

SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

The development of a residential electricity analytics framework requires a well-defined system structure that can process household consumption data and generate meaningful information for end users. An effective design should support data processing, consumption estimation, analytical evaluation, and visualization while maintaining a clear workflow between individual functional modules [1], [2].

The framework developed in this study is organized as a collection of interconnected components responsible for data preparation, feature extraction, sequence generation, prediction, analytical processing, and result presentation. Each module performs a specific task and contributes to the overall objective of transforming household electricity records into device-level consumption insights.

To support reliable operation, the system incorporates mechanisms for data handling, temporal pattern analysis, abnormal usage identification, expenditure assessment, and graphical reporting. The outputs generated by these components are combined to provide a comprehensive view of residential electricity behaviour and appliance usage characteristics.

This chapter presents the analysis and design aspects of the developed framework. It discusses the existing approaches, identified limitations, proposed solution strategy, system architecture, and operational workflow that form the foundation of the implementation described in the subsequent chapters.

3.2 Existing System

Current residential electricity monitoring approaches can generally be classified into conventional metering systems and appliance-specific monitoring solutions. Conventional metering infrastructure records cumulative household electricity consumption and provides users with information regarding overall energy usage over a specified period. While these systems are widely deployed and easy to operate, they offer limited insight into the contribution of individual appliances [1].

To obtain more detailed consumption information, appliance-specific monitoring techniques have been introduced. These approaches utilize dedicated measurement devices installed on individual appliances to capture device-level electricity usage directly. Such systems can provide detailed consumption records; however, the requirement for additional hardware increases deployment complexity and operational cost, particularly when monitoring multiple appliances simultaneously [1], [2].

Recent developments have introduced data-driven approaches capable of deriving appliance-related information from household electricity records. These methods employ analytical and learning-based techniques to separate combined consumption signals into device-level

estimates. Various architectures and computational frameworks have been explored to improve estimation accuracy and support detailed consumption analysis [2], [3], [4], [5], [6].

Although existing solutions provide useful monitoring capabilities, many implementations focus on a specific objective such as consumption measurement, device-level estimation, or performance evaluation. As a result, analytical features, expenditure assessment, abnormal usage reporting, and user-oriented visualization are often distributed across separate systems rather than being available within a single environment.

Consequently, there remains an opportunity to develop a more integrated residential electricity analysis framework capable of combining estimation, analytics, reporting, and visualization functionalities in a unified platform.

3.3 Limitations of Existing System

The key limitations observed in currently available residential electricity monitoring approaches are summarized below:

- Conventional metering systems provide only aggregate household consumption information and offer limited visibility into device-level electricity usage.
- Appliance-specific monitoring solutions require additional measurement hardware, resulting in increased deployment and maintenance costs.
- Many existing frameworks prioritize consumption estimation while providing limited analytical and decision-support capabilities.
- Identification of unusual consumption patterns and operational irregularities is often not incorporated as a core system feature.
- Expenditure assessment and appliance-wise cost reporting are frequently unavailable or implemented separately from monitoring systems.
- Several solutions provide limited support for interactive reporting and user-friendly presentation of analytical results.
- Higher model complexity may increase computational requirements and reduce deployment flexibility in resource-constrained environments.
- Existing approaches often provide limited assistance for understanding long-term consumption behaviour and improving electricity utilization.

3.4 Proposed System

- The proposed framework utilizes Non-Intrusive Load Monitoring (NILM) principles to derive device-level electricity consumption information from aggregate household energy measurements. By relying on a single input source, the framework eliminates the need for dedicated monitoring hardware on individual appliances and provides a practical solution for residential electricity analysis [1], [2].
- A Bidirectional Long Short-Term Memory (BiLSTM) network is employed as the core prediction component. Using data obtained from the UK-DALE dataset, the network learns temporal relationships within consumption records and estimates the electricity usage of selected household appliances from combined power measurements [3], [4].

- In addition to device-level consumption estimation, the framework incorporates mechanisms for identifying unusual usage behaviour and detecting significant deviations from normal consumption patterns. A cost assessment component is also included to estimate electricity expenditure based on predicted appliance usage and configurable tariff values.
- An interactive visualization environment is developed using Streamlit to present analytical outputs through charts, summaries, and monitoring panels. The interface enables users to examine consumption trends, review anomaly notifications, and evaluate estimated electricity costs through a centralized platform.
- By combining consumption estimation, anomaly analysis, expenditure assessment, and visualization capabilities within a unified framework, the proposed solution provides comprehensive insight into residential electricity usage and supports data-driven energy awareness.

3.5 Solution Strategy

- The proposed framework follows a structured workflow for transforming household electricity records into device-level consumption information. The strategy integrates data preparation, predictive modeling, analytical processing, and visualization within a unified environment.
- The process begins with residential electricity data obtained from the UK-DALE dataset. The collected records are prepared through cleaning, normalization, and temporal aggregation procedures to ensure consistency and suitability for subsequent analysis. Additional informative attributes are derived to capture temporal and consumption-related characteristics present within the data [1], [2].
- The processed records are then organized into sequential observations and supplied to a Bidirectional Long Short-Term Memory (BiLSTM) network. The model analyzes temporal relationships within the electricity data and generates consumption estimates for the selected household appliances [3], [4].
- The predicted appliance-level values are further utilized by analytical modules responsible for abnormal usage identification and electricity expenditure assessment. These modules provide additional insight into consumption behaviour and appliance operation while supporting cost-related analysis [5], [6].
- Finally, all generated outputs are consolidated within an interactive visualization platform. The interface enables users to review consumption statistics, examine analytical results, monitor unusual usage events, and assess projected electricity expenses through graphical representations and summary reports.

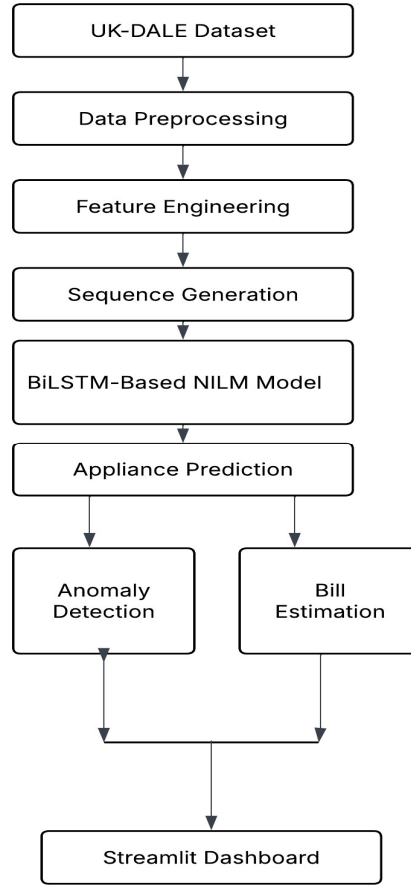


Figure 1: Overall Solution Strategy of the Proposed AI-Based Smart Energy Monitoring System

3.6 System Architecture

The architecture of the proposed residential electricity analytics framework consists of multiple interconnected modules that work together to transform household electricity data into meaningful appliance-level information. The architecture is designed to support data processing, consumption estimation, analytical evaluation, and result visualization within a unified environment.

The first component of the architecture is the data acquisition layer, which utilizes electricity consumption records obtained from the UK-DALE dataset. This layer serves as the primary source of information for all subsequent processing and analytical operations.

The collected data is forwarded to the processing layer, where data preparation and feature generation activities are performed. This layer is responsible for organizing the electricity records into a structured format suitable for predictive modeling and analysis.

The processed information is then supplied to the BiLSTM-based NILM model, which acts as the core prediction engine of the framework. The model analyzes temporal consumption patterns and generates appliance-level electricity consumption estimates for the selected household devices.

The predicted appliance consumption values are subsequently delivered to the analytical layer. This layer consists of three major modules: Appliance Analytics, Anomaly Detection, and Bill Estimation. The Appliance Analytics module provides insights into device-wise energy usage and consumption trends. The Anomaly Detection module identifies unusual consumption behaviour and significant deviations from expected operating patterns. The Bill Estimation module calculates projected electricity expenditure based on appliance-level energy consumption and user-defined tariff rates.

The outputs generated by these analytical modules are integrated into the presentation layer, which is implemented using Streamlit. This layer provides an interactive interface for monitoring energy consumption, reviewing analytical results, visualizing appliance behaviour, and examining estimated electricity costs through graphical reports and dashboards.

The interaction among these modules enables the framework to convert raw household electricity records into informative insights that support residential energy monitoring and consumption analysis. The overall architectural design of the proposed framework is illustrated in Figure 2.

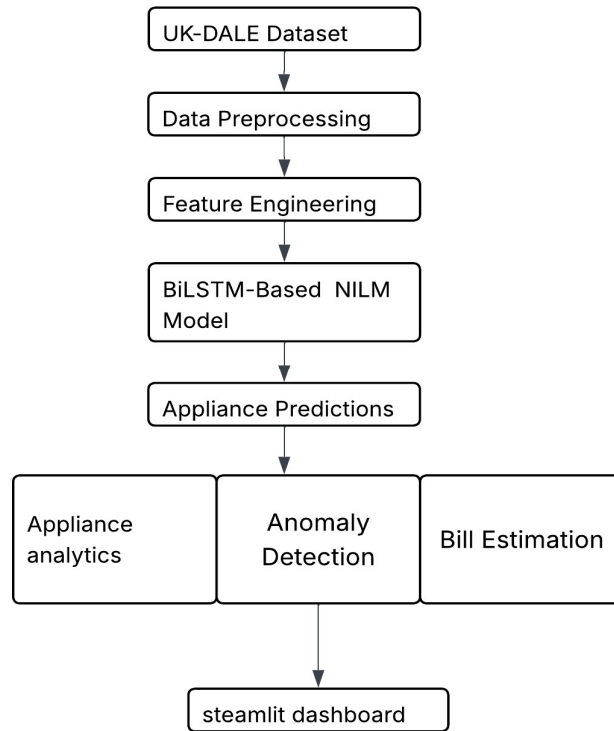


Figure 2: System Architecture of the Proposed AI-Based Smart Energy Monitoring System

3.7 Block Diagram

The block diagram illustrates the operational flow of the proposed residential electricity analytics framework and shows how electricity consumption data is processed through different stages to generate appliance-level information. Each block represents a specific function within the overall system and contributes to the transformation of raw consumption records into meaningful analytical outputs.

The process begins at the input layer, where household electricity consumption data is supplied to the framework. The collected data is first subjected to data cleaning procedures to remove inconsistencies and improve data quality. The cleaned records are then normalized to ensure a consistent numerical scale and improve learning performance.

Following normalization, the data is converted into sequential input windows through the sequence generation stage. These sequences preserve temporal relationships within the electricity records and provide suitable inputs for time-series analysis.

The generated sequences are forwarded to the BiLSTM layer, which learns consumption patterns and temporal dependencies present in the data. The extracted representations are subsequently processed by a dense layer that refines the learned information and supports appliance-level estimation.

The multi-appliance prediction block produces separate electricity consumption estimates for the selected household appliances. These predictions are then supplied to the analytics engine, which performs consumption analysis, anomaly identification, and electricity expenditure assessment.

Finally, the processed information is presented through the dashboard layer. This interface provides graphical visualizations, statistical summaries, anomaly notifications, and expenditure reports, enabling users to monitor and analyze residential electricity consumption effectively.

The detailed operational flow of the developed framework is illustrated in Figure 3.

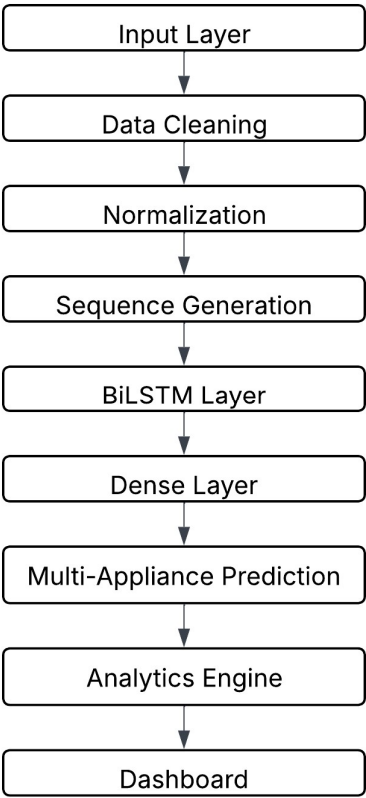


Figure 3 : Detailed Workflow of the Developed Energy Monitoring Framework

CHAPTER 4

METHODOLOGY

4.1 Introduction

This chapter outlines the procedures and techniques adopted for developing the residential energy analytics framework. The methodology defines the sequence of operations used to convert raw household electricity measurements into meaningful device-level consumption information and analytical outputs. A systematic approach was followed to ensure reliable data preparation, model development, and performance assessment.

The study utilizes the UK-DALE dataset as the primary source of residential electricity records. Before model construction, the dataset is subjected to a series of preparation steps designed to improve consistency, remove irregularities, and generate informative representations suitable for machine learning applications. These operations include data refinement, feature extraction, sequence preparation, network training, and result evaluation [1], [2].

To capture temporal characteristics present within household electricity records, a Bidirectional Long Short-Term Memory (BiLSTM) network is employed. The learning process incorporates multiple informative attributes derived from historical consumption behaviour and time-related patterns, enabling the network to recognize complex relationships within the data [3], [4].

The outputs generated by the prediction network are subsequently utilized for consumption analysis, abnormal pattern identification, and electricity cost assessment. The resulting information is presented through an interactive visualization environment that supports exploration of usage behaviour and monitoring of household electricity trends. The methodology adopted in this work encompasses data preparation, feature extraction, sequence modelling, network training, prediction generation, and performance analysis stages [5], [6].

4.2 Dataset Description

The experimental study conducted in this work is based on the UK Domestic Appliance-Level Electricity (UK-DALE) dataset, a publicly available collection of residential electricity measurements frequently adopted for energy disaggregation research. The dataset contains information recorded from real households and includes both overall household electricity readings and device-specific consumption records, making it suitable for supervised learning applications related to residential energy analysis [7].

For model development, data corresponding to Building 1 of the dataset is utilized. Five household devices are selected for investigation: Refrigerator, Washing Machine, Dishwasher, Television, and Kettle. These appliances represent different usage behaviours, ranging from continuously operating equipment to devices that exhibit short-duration and event-driven activity patterns. Such diversity enables the learning framework to capture a broad range of electricity usage characteristics.

The original measurements are collected over an extended observation period and contain high-resolution electricity records. To improve computational efficiency and simplify sequence-based learning, the measurements are aggregated into hourly intervals. This transformation reduces data volume while preserving the overall consumption behaviour required for model development and evaluation.

Within the learning framework, the combined household electricity profile serves as the primary input, whereas the corresponding device-specific readings are used as reference outputs during training and testing. The availability of these reference measurements enables the network to learn the relationship between total household demand and the contribution of individual appliances.

Due to its comprehensive appliance coverage, real-world residential measurements, and extensive use in academic studies, the UK-DALE dataset provides a reliable benchmark for assessing the effectiveness of the developed energy analytics framework [7].

Sl. No.	Appliance
1	Refrigerator
2	Washing Machine
3	Dishwasher
4	Television
5	Kettle

Table 3. Selected Appliances from the UK-DALE Dataset

4.3 Data Pre-processing

The effectiveness of a deep learning model is strongly influenced by the quality and consistency of the data supplied during training. Therefore, a dedicated preparation stage is performed before model development to ensure that the electricity records are suitable for sequence-based learning. The objective of this stage is to refine the raw measurements, eliminate inconsistencies, and create a reliable dataset for subsequent analysis.

The initial step involves selecting the appliance channels required for the study. Five household devices—Refrigerator, Washing Machine, Dishwasher, Television, and Kettle—are considered for consumption estimation. Any incomplete observations or unreliable entries are addressed during this stage to improve data integrity. In cases where overall household demand is not explicitly available, it is derived by combining the electricity usage of the selected devices [7].

Since the original measurements contain a large number of observations, temporal aggregation is performed to improve computational efficiency. The recorded values are consolidated into hourly intervals using energy-based summation, ensuring that the total electricity usage within each hour remains unchanged. This transformation reduces data volume while preserving the broader consumption behaviour required for model learning.

After temporal aggregation, the dataset is separated into training and testing portions using a chronological division strategy. To ensure numerical consistency among the input variables,

Min-Max scaling is applied, transforming the feature values into a standardized range. This step contributes to faster convergence and more stable learning during network optimization.

The prepared dataset serves as the basis for the subsequent feature extraction and sequence construction stages. Through these operations, the electricity records become more suitable for pattern recognition, enabling the learning framework to capture underlying relationships present within residential consumption data. The complete preprocessing sequence adopted in this study is illustrated in Figure 4.

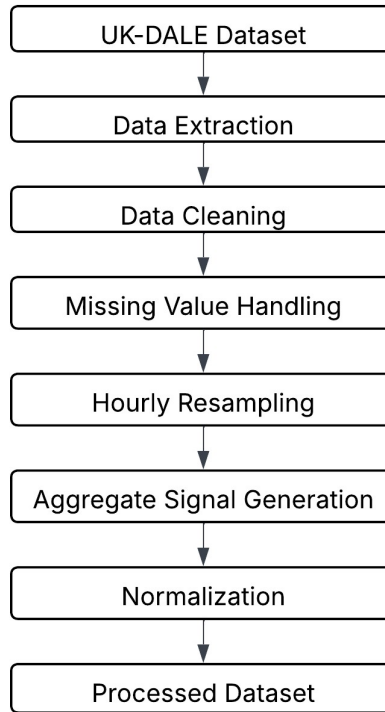


Figure 4 : Data Pre-processing Workflow

4.4 Feature Engineering

The raw electricity measurements obtained after data preparation contain limited contextual information for identifying appliance operating characteristics. To improve the learning capability of the prediction network, additional descriptive attributes are generated from the household consumption records. This process enriches the input representation and enables the network to recognize recurring usage behaviour more effectively.

Several time-related attributes are derived from the timestamp associated with each observation. These include indicators representing the hour, day, and month of occurrence. A separate weekend flag is also introduced to distinguish between weekday and weekend activity patterns, as residential electricity usage frequently varies across different periods of the week.

To provide historical context, previous consumption values are incorporated into the feature set. These lag variables enable the network to consider recent behaviour when estimating future appliance activity. In addition, moving-average statistics calculated over different observation windows are included to capture short-term trends and smooth fluctuations present within the electricity records.

Periodic characteristics of time are represented through sine and cosine transformations. These cyclical encodings allow the learning framework to recognize repeating daily, weekly, and seasonal patterns without introducing artificial discontinuities between adjacent time periods [3], [4].

The resulting feature collection combines consumption measurements, temporal indicators, historical variables, statistical summaries, and cyclical encodings. This enriched representation provides the information required for subsequent sequence construction and network training stages.

4.5 Sequence Generation

After feature extraction, the processed electricity consumption records are transformed into sequential samples suitable for time-series learning. Since household energy usage exhibits temporal dependencies, the prediction model requires historical observations rather than individual data points. Therefore, a sequence generation procedure is applied to preserve the chronological relationship between consecutive measurements.

A sliding-window approach is utilized to create fixed-length input sequences from the processed dataset. Each sequence consists of 24 consecutive hourly observations, representing one day of household electricity consumption history. The generated sequences contain the aggregate household consumption signal along with the engineered features derived during the previous stage. This arrangement enables the learning framework to analyze both historical energy usage patterns and temporal characteristics simultaneously.

For every input sequence, the corresponding target output consists of appliance-level electricity consumption values for the selected appliances, namely Refrigerator, Washing Machine, Dishwasher, Television, and Kettle. Thus, the generated dataset contains paired input-output samples that can be used for supervised learning.

The sequence generation process converts the tabular dataset into a three-dimensional structure suitable for the Bidirectional Long Short-Term Memory (BiLSTM) network. The resulting input data is represented in the form: (Number of Samples $\times$ Sequence Length $\times$ Number of Features)

where the sequence length is fixed at 24 hours and the feature dimension corresponds to the engineered attributes generated during the feature engineering stage.

By preserving temporal continuity and historical consumption behaviour, the sequence generation stage enables the BiLSTM network to learn long-term dependencies present in household electricity data. The generated sequences form the final input to the prediction model and provide the foundation for appliance-level energy estimation in the subsequent stage.

4.6 BiLSTM-Based NILM Model

The prediction framework developed in this study is based on a Bidirectional Long Short-Term Memory (BiLSTM) network. This architecture was selected because residential electricity usage exhibits strong sequential characteristics, requiring a learning mechanism capable of utilizing information from multiple time steps. By processing input sequences in both temporal directions, the network can extract richer contextual information and improve its ability to identify relationships embedded within household electricity records [4], [5].

The network receives sequential samples generated during the sequence construction stage. Each sample contains 24 hourly observations together with the descriptive attributes produced during feature extraction. These inputs provide information related to historical consumption behaviour, temporal patterns, and recurring usage cycles, enabling the network to learn complex relationships from the available data.

The implemented architecture contains two stacked Bidirectional LSTM layers with 128 and 64 hidden units respectively. To stabilize the learning process, Batch Normalization is applied after each recurrent layer. In addition, Dropout regularization with a probability of 0.2 is incorporated to minimize excessive dependence on specific training samples and improve generalization performance [3], [4].

The output generated by the recurrent layers is forwarded to a fully connected layer consisting of 64 neurons. L2 regularization with a coefficient of 0.0001 is incorporated within this stage to improve robustness during training. Instead of producing a single prediction, the architecture employs multiple output branches corresponding to Refrigerator, Washing Machine, Dishwasher, Television, and Kettle consumption estimates. This arrangement enables simultaneous learning for multiple appliances while preserving appliance-specific prediction capabilities.

Network optimization is performed using the Huber Loss function. This objective function provides a balance between absolute-error and squared-error minimization, making it suitable for electricity datasets that may contain sudden consumption variations and irregular operating events [5], [6].

The trained network acts as the central prediction component of the developed framework. The generated estimates are subsequently utilized by the analytical, cost evaluation, and visualization modules, enabling the extraction of meaningful information from household electricity measurements. The detailed configuration of the network is presented in Table 4.

Parameter	Value
Sequence Length	24
Number of Features	13
BiLSTM Layer 1	128 Units
BiLSTM Layer 2	64 Units
Batch Normalization	2 Layers
Dense Layer	64 Units
Dropout Rate	0.2
L2 Regularization	0.0001
Output Appliances	5

Table 4. BiLSTM Model Configuration

4.7 Model Training

Once the sequential samples were prepared, the learning framework was trained to establish the relationship between household electricity measurements and the corresponding device-specific consumption values. To maintain the temporal nature of the dataset, the generated samples were divided chronologically, with 80% allocated for learning and the remaining 20% reserved for performance assessment. This strategy ensured that future observations were excluded from the learning phase and preserved the integrity of time-series evaluation.

Each training sample consisted of a 24-hour observation window together with the associated appliance targets. During optimization, the network repeatedly processed these samples and refined its internal parameters to improve prediction accuracy. The Adam optimization algorithm was selected with a learning rate of 0.001 to facilitate efficient parameter updates and stable convergence throughout the learning process.

To improve robustness against irregular consumption events, the Huber Loss function was adopted as the optimization criterion. Additional regularization measures, including Dropout and L2 penalties, were incorporated to reduce excessive model specialization and enhance performance on previously unseen data.

A validation subset representing 15% of the training data was utilized to monitor learning progress during optimization. Two adaptive training mechanisms were also employed. The first, Early Stopping, automatically terminated the learning process when further improvement became negligible. The second, ReduceLROnPlateau, gradually lowered the learning rate whenever validation performance stagnated, enabling finer parameter adjustments.

Training was conducted using a batch size of 64 with an upper limit of 200 epochs. Throughout this process, optimization and validation losses were continuously observed to assess learning behaviour and verify model stability. The resulting learning curves provided evidence regarding convergence characteristics and overall network performance.

After completion of training, the finalized network was stored in Keras format under the filename *nilm_ukdale_v3.keras*. This trained model was subsequently incorporated into the analytical framework for consumption estimation, abnormal usage assessment, expenditure calculation, and visualization tasks. The complete set of training parameters is presented in Table 5.

Parameter	Value
Optimizer	Adam
Learning Rate	0.001
Loss Function	Huber Loss
Batch Size	64
Epochs	200
Validation Split	15%
Early Stopping Patience	15
LR Reduction Patience	6
Training-Test Split	80:20

Table 5. Training Parameters of the Proposed Model

4.8 Performance Evaluation Metrics

The effectiveness of the developed prediction framework is assessed using quantitative measures that evaluate both consumption estimation quality and appliance operating-state recognition. These metrics provide objective evidence of model performance and enable comparison with simpler reference approaches. The selected measures include Mean Absolute Error (MAE), Precision, Recall, F1 Score, and Baseline Comparison.

4.8.1 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) quantifies the average magnitude of prediction errors by measuring the absolute difference between actual and estimated consumption values. Smaller MAE values indicate closer agreement between predicted outputs and reference measurements.

$$[MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|]$$

where:

(y_i) = Actual consumption value

$(\hat{y}_i)$ = Predicted consumption value

(N) = Total number of observations

4.8.2 Precision

Precision evaluates the reliability of appliance activation predictions. It represents the proportion of correctly identified ON states among all instances predicted as ON.

$$[\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}]$$

where:

TP = True Positives

FP = False Positives

4.8.3 Recall

Recall measures the ability of the model to identify actual appliance activation events. It indicates the proportion of true ON states that are successfully detected.

$$[\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}]$$

where:

FN = False Negatives

4.8.4 F1 Score

The F1 Score combines Precision and Recall into a single indicator and provides a balanced measure of classification effectiveness. Higher values indicate better operating-state recognition performance.

$$[\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}]$$

4.8.5 Naive Baseline Comparison

To determine the effectiveness of the developed network, its performance is compared with a simple reference approach. The baseline method estimates appliance consumption by allocating household electricity usage according to historical average appliance proportions. Comparing both approaches helps quantify the benefits achieved through deep learning-based prediction.

The combination of MAE, Precision, Recall, F1 Score, and Baseline Comparison provides a comprehensive assessment of estimation accuracy and appliance state recognition capability.

The combination of MAE, Precision, Recall, F1 Score, and Baseline Comparison provides a comprehensive evaluation of both regression performance and appliance state detection capability, enabling a detailed assessment of the proposed Smart Energy Monitoring System.

Chapter 5

IMPLEMENTATION

5.1 Introduction

This chapter describes the practical realization of the residential energy analytics framework developed in this study. The objective of the implementation phase is to convert the conceptual design and methodological procedures presented in previous chapters into an operational software system capable of processing electricity consumption records and generating meaningful analytical outputs.

The framework is developed using Python and a collection of open-source tools that support data handling, machine learning, visualization, and application development. Residential electricity measurements obtained from the UK-DALE dataset are utilized throughout the implementation process, while a Bidirectional Long Short-Term Memory (BiLSTM) network serves as the primary prediction engine responsible for estimating device-wise electricity usage.

The software solution is organized into multiple functional components that collectively perform data preparation, feature extraction, sequence construction, network training, prediction generation, performance assessment, and analytical processing. Additional modules are incorporated to support abnormal usage identification, electricity expenditure calculation, and consumption analysis.

To improve accessibility and user interaction, a web-based visualization environment is developed using Streamlit. This interface presents graphical summaries, consumption trends, appliance estimates, alert notifications, and cost-related information through an interactive dashboard.

The following sections discuss the software tools, implementation modules, network realization, evaluation procedures, and visualization features incorporated within the developed framework.

5.2 Development Environment

The implementation of the developed framework is carried out using Python, which provides a flexible environment for scientific computing, machine learning, and application development. A collection of open-source libraries and software tools is utilized to support different stages of development, ranging from data handling and model construction to visualization and deployment.

TensorFlow and Keras are employed for building and training the deep learning network used in this study. These libraries provide high-level functionalities for defining neural network architectures, optimizing model parameters, and managing the training process. Data preparation and numerical computations are performed using Pandas and NumPy, which offer efficient mechanisms for handling large-scale time-series datasets and mathematical operations.

Scikit-Learn is incorporated for feature scaling and performance assessment, while Matplotlib is used to generate graphs, learning curves, and analytical visualizations. These tools assist in interpreting model behaviour and presenting experimental findings in a clear manner.

A web-based interface is developed using Streamlit to provide interactive access to analytical results. The dashboard enables visualization of consumption trends, prediction outputs, abnormal usage notifications, and expenditure-related information through an intuitive user interface.

The UK-DALE dataset serves as the primary data source for network training and evaluation. The software tools and technologies employed during implementation are summarized in Table 6.

Sl. No.	Tool / Technology	Purpose
1	Python	Programming Language
2	TensorFlow / Keras	BiLSTM Model Development
3	Pandas	Data Processing and Analysis
4	NumPy	Numerical Computation
5	Scikit-Learn	Evaluation Metrics
6	Matplotlib	Data Visualization
7	Streamlit	Dashboard Development
8	UK-DALE Dataset	Energy Consumption Dataset

Table 6. Software Tools and Technologies Used

5.3 Dataset Preparation Module

The Dataset Preparation Module is responsible for transforming the raw UK-DALE dataset into a structured format suitable for appliance-level energy disaggregation. This module performs the initial data processing tasks required before model training and evaluation.

The process begins by loading household electricity consumption data and selecting the appliances considered in this study, namely Refrigerator, Washing Machine, Dishwasher, Television, and Kettle. The corresponding appliance-level energy measurements are extracted and organized into a unified dataset. Missing values and incomplete records are handled appropriately to ensure data consistency and reliability.

To improve computational efficiency, the energy consumption data is resampled into hourly intervals while preserving overall energy usage characteristics. Aggregate household consumption is then generated by combining the energy usage of the selected appliances. This aggregate signal serves as the primary input to the NILM model, while the individual appliance measurements are retained as target outputs for supervised learning.

The prepared dataset is further organized into a structured format that supports feature engineering, sequence generation, model training, and performance evaluation. By ensuring data quality and consistency, the Dataset Preparation Module provides a reliable foundation for the subsequent stages of the proposed Smart Energy Monitoring System.

5.4 BiLSTM Model Implementation

The Dataset Preparation Module serves as the initial processing component of the implemented framework. Its primary function is to organize the raw UK-DALE records into a structured format suitable for subsequent analytical and predictive operations.

The module begins by importing the required household electricity records and selecting the appliance channels considered in the study. Measurements corresponding to Refrigerator, Washing Machine, Dishwasher, Television, and Kettle are extracted and consolidated into a unified data structure. Data validation procedures are performed to identify incomplete entries and maintain consistency throughout the dataset.

To improve processing efficiency and reduce data volume, the recorded measurements are aggregated into hourly intervals. This transformation preserves the overall consumption behaviour while producing a more manageable dataset for model development. A combined household electricity profile is subsequently derived from the selected appliance records and utilized as the primary input signal for the prediction framework.

Following data organization, the processed records are stored in a format that supports feature extraction, sequence construction, network training, and result evaluation. The module therefore acts as a bridge between the raw dataset and the learning framework, ensuring that the information supplied to subsequent components is consistent, reliable, and suitable for analysis.

The successful execution of this module establishes the data foundation required for the remaining stages of the implemented system.

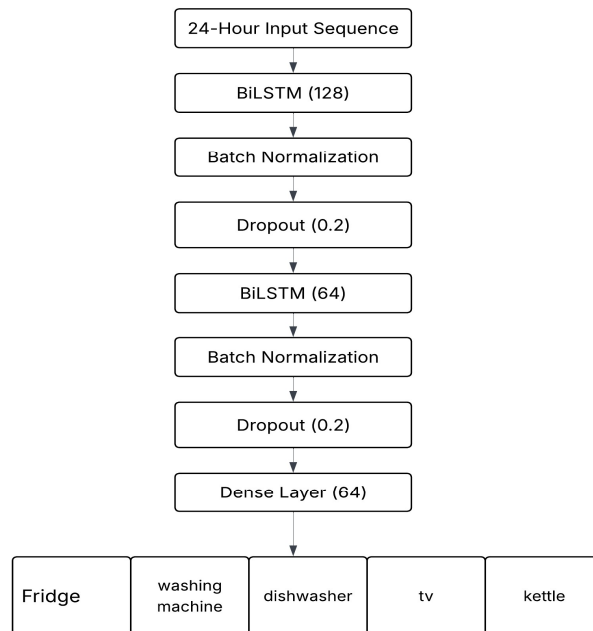


Figure 5 : BiLSTM Model Implementation Architecture

5.5 Evaluation Module

The Evaluation Module is designed to examine the effectiveness of the trained prediction network and generate performance reports for subsequent analysis. Once prediction results are produced, the module compares the estimated appliance values with the corresponding reference measurements obtained from the dataset. This process enables a systematic assessment of the network's estimation capability and operating-state recognition performance.

The module computes a collection of statistical indicators that quantify prediction quality and classification effectiveness. These measures are automatically generated after inference and provide numerical evidence regarding the accuracy and reliability of the developed framework. The calculated results are subsequently organized into structured reports for interpretation and comparison.

To provide additional insight into appliance operating-state recognition, the module generates confusion matrices for each monitored device. These visual representations summarize the distribution of correct and incorrect classifications and assist in identifying patterns associated with prediction errors. Appliance-specific operating thresholds are also determined to facilitate ON/OFF state analysis.

A comparative assessment is further performed using a simple reference approach that estimates appliance usage based on historical consumption proportions. The comparison enables the measurement of performance improvements achieved by the trained deep learning network relative to a conventional estimation technique.

The outputs generated by this module include performance statistics, comparative analyses, and visualization artifacts that support detailed examination of network behaviour. These results form the basis for evaluating the effectiveness of the implemented framework and validating its suitability for residential electricity monitoring applications.

5.6 Streamlit Dashboard Implementation

A web-based dashboard is developed using Streamlit to provide an interactive environment for exploring the outputs generated by the implemented framework. The dashboard acts as the presentation layer of the application, allowing users to access analytical results, visual summaries, and system-generated insights through a unified interface.

The interface is organized into several functional sections, each designed to support a specific aspect of residential electricity analysis. These sections include Live Monitor, Appliance Analysis, Anomaly Detection, Bill Estimator, and Live Simulation. Together, they provide convenient access to consumption statistics, prediction results, operational insights, and expenditure-related information.

The Live Monitor section presents key household electricity indicators such as average daily usage, peak demand, projected monthly consumption, and estimated costs. Interactive charts are incorporated to display historical trends and facilitate comparison between estimated and reference values.

The Appliance Analysis section focuses on device-specific information by presenting consumption statistics, performance indicators, and usage trends. The Anomaly Detection section highlights unusual operating patterns that deviate from normal behaviour, while the Bill Estimator section calculates expected electricity expenses using user-defined tariff rates. In addition, the Live Simulation section allows users to observe how changes in appliance activity influence overall household electricity usage.

A variety of graphical elements, including trend charts, comparison plots, summary indicators, and statistical visualizations, are incorporated to improve interpretability and enhance user engagement. These visual components assist users in examining consumption behaviour and understanding the impact of different appliances on overall electricity demand.

By consolidating analytical outputs, performance information, expenditure estimates, and visualization tools within a single environment, the dashboard provides a practical interface for monitoring and interpreting residential electricity usage. The implemented dashboard is illustrated in Figure 6.

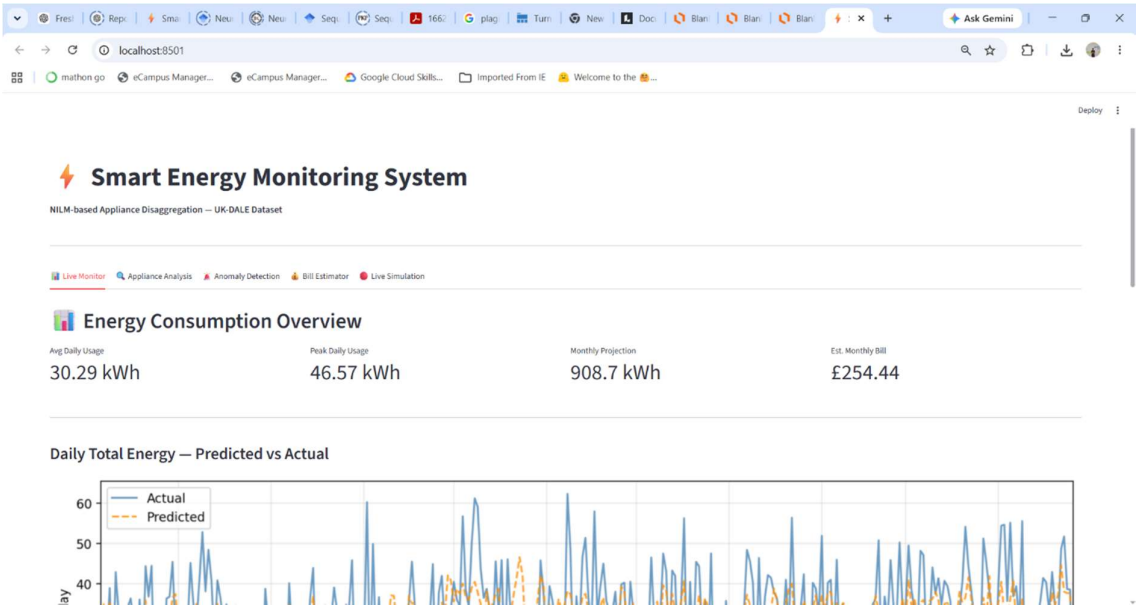


Figure 6 : User Interface of the Streamlit-Based Energy Monitoring Dashboard

Chapter 6

RESULTS AND DISCUSSION

6.1 Introduction

This chapter presents the experimental findings obtained from the developed residential energy analytics framework and examines the effectiveness of the trained prediction network. The analysis is performed using the testing portion of the UK-DALE dataset to investigate the ability of the framework to infer device-specific electricity usage from combined household consumption records.

A detailed assessment is carried out using both regression-based and classification-based performance measures. Statistical indicators, comparative studies, confusion matrix analysis, and graphical visualizations are utilized to examine prediction behaviour and evaluate the reliability of the generated outputs. Performance comparisons with a reference approach are also included to quantify the benefits achieved through the adopted learning framework.

In addition to numerical evaluation, the chapter presents the outputs generated by the visualization interface, including consumption trends, expenditure analysis, abnormal usage notifications, and device-specific insights. These results provide evidence regarding the practical applicability of the developed framework in residential electricity analysis.

The following sections discuss network performance, prediction outcomes, comparative evaluation, visualization results, and key observations derived from the experimental study.

6.2 Model Training Performance

The learning behaviour of the trained network was examined using the optimization and validation loss curves obtained during training. Huber Loss was monitored throughout the learning process to evaluate convergence characteristics and assess the stability of parameter optimization.

As illustrated in Figure 7, both curves exhibit a rapid decline during the initial training epochs, indicating effective learning from the input sequences. The reduction in loss values demonstrates that the network successfully captures meaningful patterns from the electricity consumption records. The validation curve reaches its lowest value at approximately Epoch 13, representing the point at which the network achieved its best generalization performance.

Beyond this stage, the optimization loss continues to decrease gradually, whereas the validation curve remains relatively stable with only minor variations. This behaviour suggests limited overfitting and indicates that the regularization mechanisms incorporated within the architecture effectively control excessive specialization. The combined use of Dropout, L2 regularization, Early Stopping, and adaptive learning-rate adjustment contributed to maintaining stable learning behaviour throughout the training process.

The observed learning curves confirm that the network converged successfully and maintained consistent performance on previously unseen data. The close agreement between optimization

and validation trends further indicates that the trained model achieved a satisfactory balance between learning capability and generalization performance.

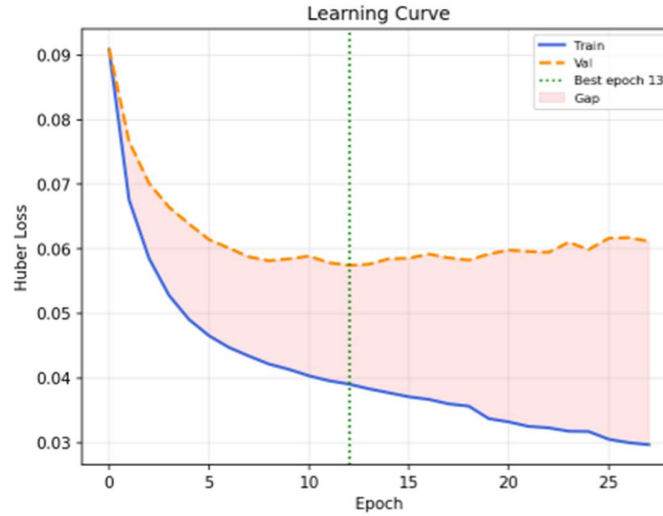


Figure 7 : Training and Validation Learning Curve of the Proposed BiLSTM Model

6.3 Appliance-Level Prediction Results

The prediction outputs generated during testing were examined for five household devices: Refrigerator, Washing Machine, Dishwasher, Television, and Kettle. Performance was assessed by comparing the estimated values with the corresponding reference measurements available in the dataset. Figures 8(a)–8(e) present the comparison between actual and estimated consumption profiles for each appliance.

The results reveal noticeable differences in prediction quality among the evaluated appliances. Such variations are primarily influenced by the operational characteristics of individual devices, including usage frequency, cycle duration, and the distinctiveness of their electricity demand patterns.

The Refrigerator and Washing Machine recorded comparatively lower state-recognition scores, with F1 values of 0.138 and 0.147 respectively. These devices typically exhibit variable operating cycles and consumption behaviour that overlaps with other household loads. Consequently, accurately identifying all activation events remains a challenging task, although the overall usage trend is still partially captured by the network.

The Dishwasher achieved moderate performance, obtaining an F1 Score of 0.482 with a Mean Absolute Error of 0.1073 kWh. The estimated profile successfully reflects the major operating cycles of the appliance, indicating that a substantial portion of its consumption behaviour was learned during training.

Among the continuously used appliances, Television exhibited strong performance with an F1 Score of 0.824 and an MAE of 0.1598 kWh. The close correspondence between estimated and reference values demonstrates effective recognition of usage periods and operating states.

The highest performance was observed for the Kettle, which achieved an F1 Score of 0.996 and an MAE of 0.1463 kWh. Since kettle operation is characterized by short-duration, high-intensity consumption events, the resulting pattern is comparatively easier to distinguish from other household loads. This contributed to highly accurate state recognition and estimation results.

Overall, the experimental observations indicate that prediction quality improves for appliances exhibiting clear and distinctive operating signatures. Devices with irregular behaviour and overlapping demand characteristics remain more challenging to estimate accurately. Nevertheless, the obtained results demonstrate the capability of the developed framework to extract meaningful device-level information from household electricity measurements.

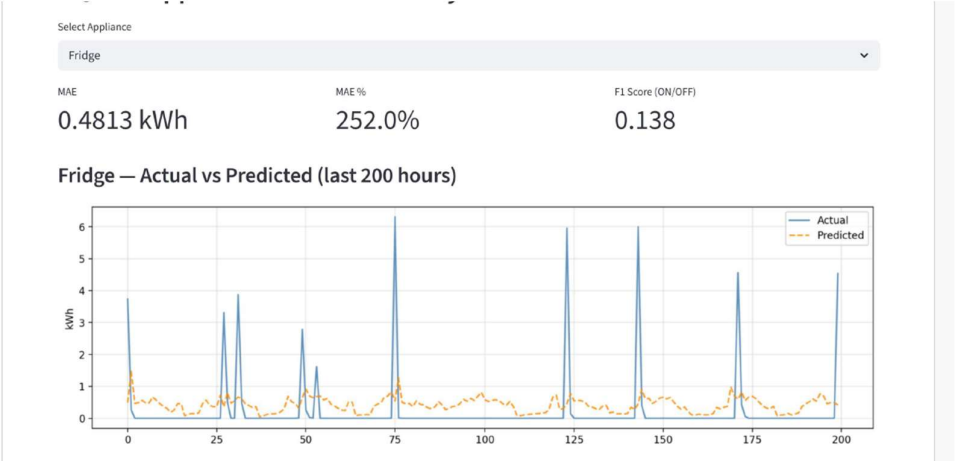


Figure 8 : Appliance-Level Energy Prediction Results

Figure 8(a) : Fridge – actual vs predicted

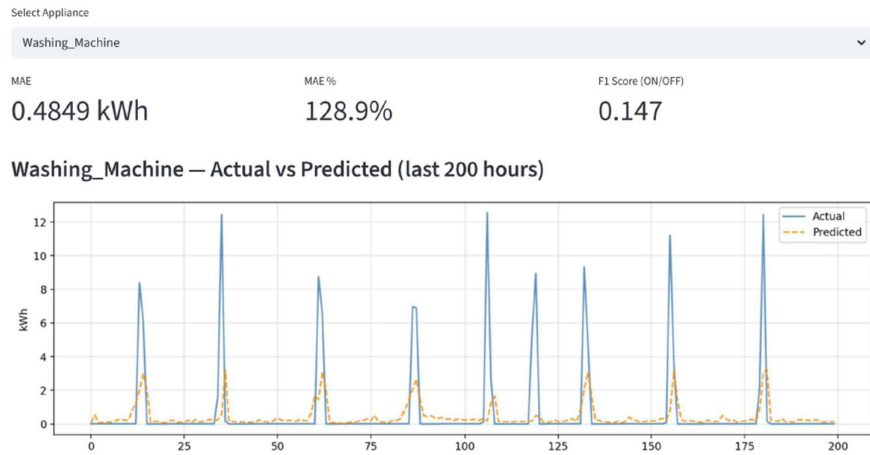


Figure 8(b) : Washing Machine – actual vs predicted

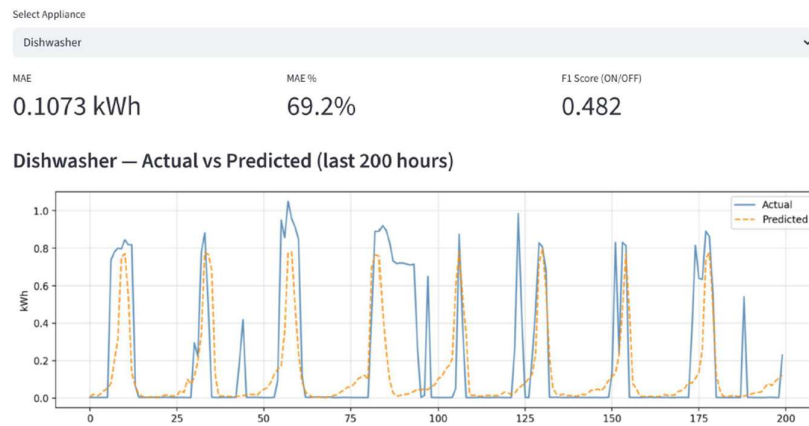


Figure 8(c) : Dishwasher – actual vs predicted

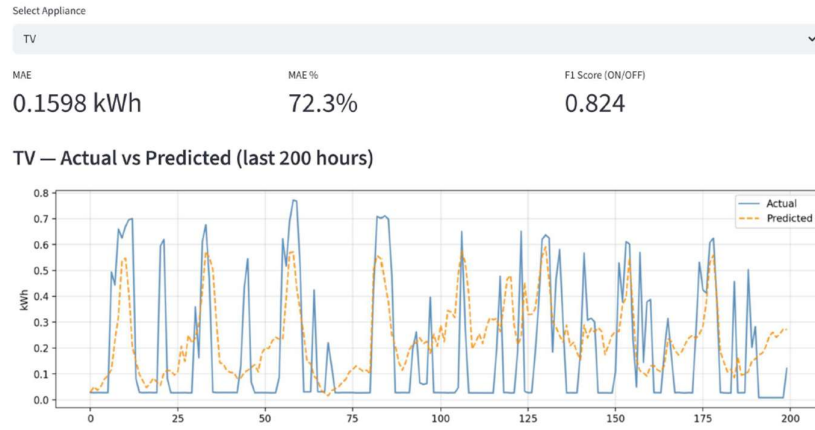


Figure 8(d) : TV – actual vs predicted

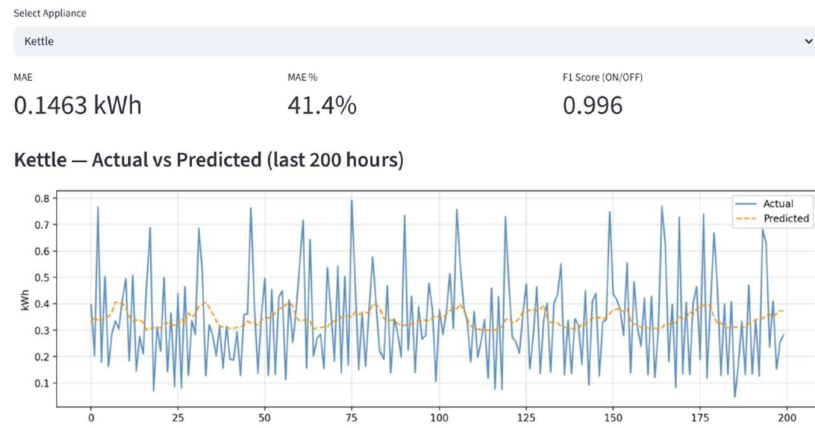


Figure 8(e) : Kettle – actual vs predicted

6.4 Performance Evaluation Results

The effectiveness of the developed prediction framework was assessed using Mean Absolute Error (MAE) and F1 Score. These measures provide complementary perspectives on performance by evaluating both numerical estimation accuracy and appliance operating-state recognition capability. The appliance-wise results obtained during testing are presented in Table 7.

The evaluation results indicate that performance varies according to the operational characteristics of individual appliances. Devices exhibiting distinct and easily distinguishable consumption signatures generally achieved higher state-recognition scores, whereas appliances with variable operating cycles and overlapping demand characteristics produced comparatively lower results.

Among the evaluated appliances, the Kettle recorded the highest F1 Score of 0.996, demonstrating excellent identification of operating events. Television also achieved strong classification performance with an F1 Score of 0.824, indicating reliable recognition of usage periods. These results suggest that appliances with well-defined consumption patterns are easier for the network to distinguish from the combined household load profile.

Moderate performance was observed for the Dishwasher, which achieved an F1 Score of 0.482. Although several operating events were successfully identified, certain usage periods remained difficult to distinguish with complete accuracy. In contrast, the Refrigerator and Washing Machine produced lower F1 Scores of 0.138 and 0.147 respectively, reflecting the challenges associated with devices that exhibit irregular operating behaviour and overlapping consumption characteristics.

From a regression perspective, the Dishwasher achieved the lowest MAE value of 0.1073 kWh, indicating the smallest average estimation error among the evaluated appliances. The remaining devices also maintained acceptable error levels, demonstrating the ability of the framework to approximate appliance consumption trends despite differences in classification performance.

Overall, the evaluation results indicate that the developed framework performs most effectively for appliances with distinctive electricity usage patterns while maintaining useful estimation capability across all monitored devices. The obtained metrics demonstrate the suitability of the approach for extracting meaningful device-level information from residential electricity data.

Appliance	MAE (kWh)	F1 Score
Refrigerator	0.4813	0.138
Washing Machine	0.4849	0.147
Dishwasher	0.1073	0.482
Television	0.1598	0.824
Kettle	0.1463	0.996

Table 7. Appliance-Wise Performance Evaluation Results

6.4.1 On/Off Detection Performance

The operating-state recognition capability of the developed framework was further examined using confusion matrix analysis. Confusion matrices provide a detailed representation of classification outcomes by comparing actual appliance states with the corresponding predicted states. This analysis enables the identification of correctly recognized activation events as well as instances of misclassification.

Figures 9(a)–9(e) illustrate the confusion matrices obtained for the selected household appliances. The distribution of values within these matrices provides insight into the ability of

the network to distinguish between active and inactive operating conditions. A larger concentration of values along the main diagonal indicates stronger classification performance, whereas higher off-diagonal values represent incorrect state assignments.

The obtained matrices reveal that classification performance differs among appliances due to variations in operating behaviour and electricity usage characteristics. Appliances exhibiting clear and distinguishable consumption patterns generally produce a higher number of correctly identified states. Conversely, devices with repetitive cycles, fluctuating demand, or consumption profiles that resemble other household loads tend to generate more classification errors.

For appliances such as Television and Kettle, the confusion matrices show a strong concentration of correctly classified ON and OFF states, indicating reliable operating-state recognition. In contrast, Refrigerator and Washing Machine display a greater number of misclassified instances, reflecting the increased difficulty associated with identifying their operating conditions from combined household electricity measurements. Dishwasher demonstrates intermediate performance, with a reasonable balance between correctly identified events and classification errors.

Overall, the confusion matrix analysis supports the quantitative results presented earlier and provides additional insight into the strengths and limitations of the developed framework. The findings indicate that appliances with distinctive operating signatures are recognized more accurately, while devices exhibiting complex consumption behaviour remain comparatively challenging to classify.

Figure 9 : On/Off Detection Confusion Matrix

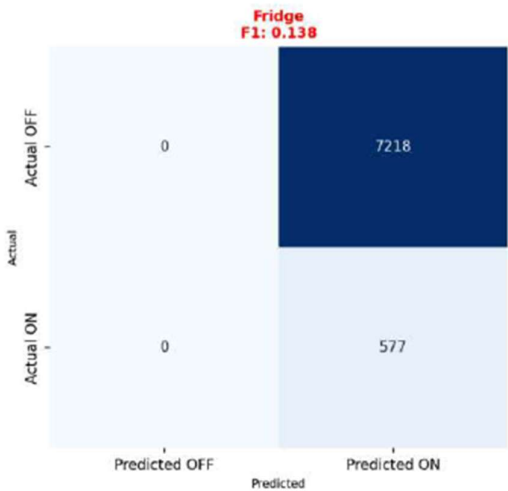


Figure 9 (a) : For Fridge ON/OFF Detection

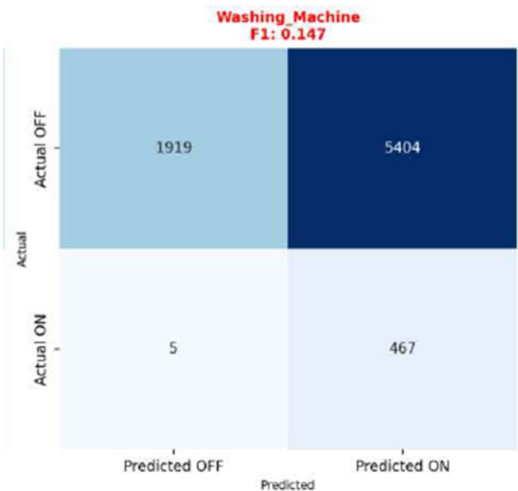
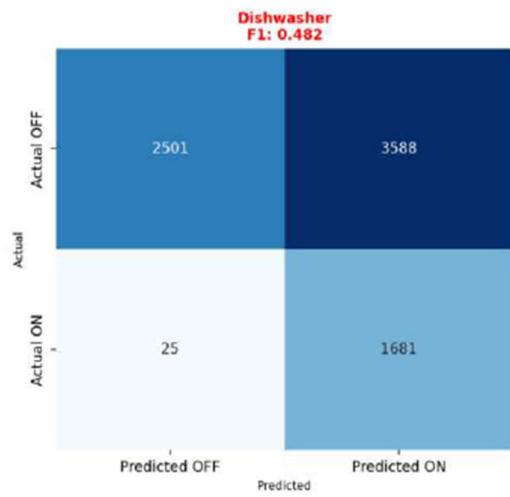
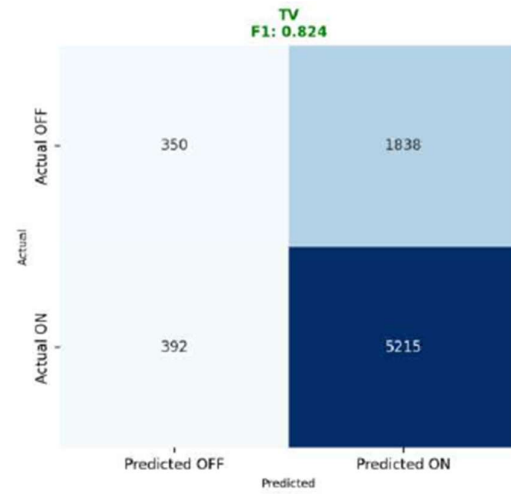


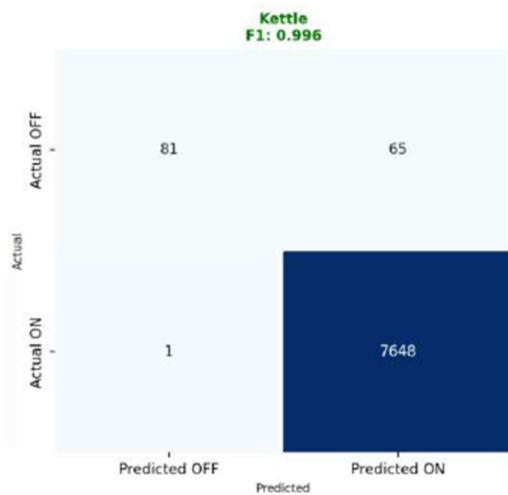
Figure 9 (b) : For Washing Machine ON/OFF Detection



**Figure 9 (c) : For Dishwasher
ON/OFF Detection**



**Figure 9 (d) : For TV ON/OFF
Detection**



**Figure 9 (e) : For Kettle
ON/OFF Detection**

6.5 Dashboard Results

The visualization interface provides a consolidated view of the analytical outputs generated during testing and enables effective interpretation of household electricity usage patterns. Through interactive charts, summary indicators, and statistical reports, users can explore consumption behaviour and examine the contribution of individual appliances to overall electricity demand.

Figure 6.4 presents a comparison between reference and estimated daily electricity usage. The observed trends indicate a strong correspondence between the two profiles, demonstrating that the prediction framework successfully captures the general variations in household demand over time. Although minor differences are visible during certain intervals, the overall consumption behaviour is represented effectively.

The appliance-wise energy distribution shown in Figure 6.5 provides insight into the relative contribution of each monitored device. The results indicate that the Refrigerator accounts for the largest proportion of total consumption at approximately 28.9%, followed by the Kettle (27.4%), Washing Machine (19.6%), Television (14.3%), and Dishwasher (9.9%). Such information can assist users in identifying the appliances responsible for the majority of household electricity usage.

Figure 6.6 illustrates daily and monthly consumption statistics for individual appliances. The generated summaries reveal that the Refrigerator and Kettle contribute the highest average usage levels, whereas the Dishwasher represents the smallest share among the evaluated devices. These observations help users understand long-term consumption behaviour and identify areas where efficiency improvements may be possible.

In addition to consumption statistics, the interface provides indicators such as average daily usage, peak demand, projected monthly consumption, and estimated expenditure. The combination of numerical summaries and graphical representations improves the accessibility of analytical results and supports informed decision-making regarding electricity utilization.

Overall, the presented outputs demonstrate the effectiveness of the visualization interface in transforming prediction results into understandable and actionable information for household electricity analysis.

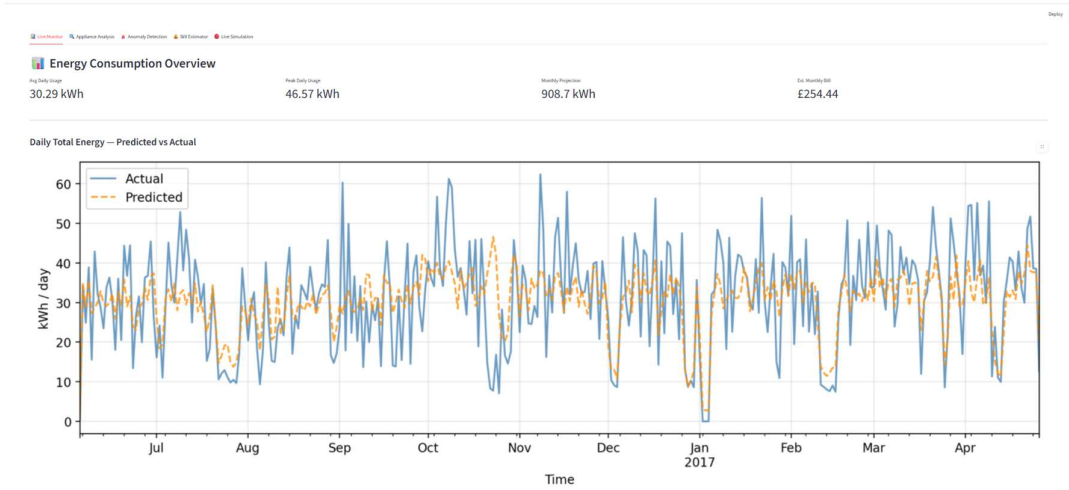


Figure 10 : Daily Total Energy Predicted v/s Actual

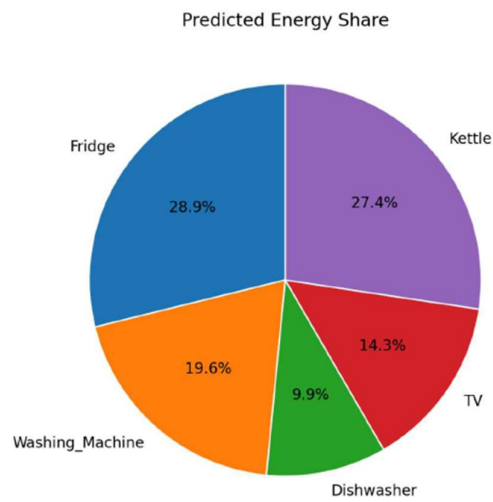


Figure 11 : Predicted Energy Share

Per-Appliance Daily Average

Appliance	Avg Daily kWh	Monthly kWh
Fridge	8.74	262.191
Washing_Machine	5.939	178.157
Dishwasher	3.002	90.054
TV	4.318	129.547
Kettle	8.293	248.776

Figure 12 : Per- Appliance Daily Average

6.6 Anomaly Detection Results

An analysis of unusual electricity usage behaviour was performed to identify significant deviations from normal consumption trends. The implemented detection mechanism utilizes a rolling 7-day Z-score approach to examine variations in appliance activity and highlight events that differ substantially from expected operating patterns.

Figure 13 illustrates the anomaly analysis results obtained during the evaluation period. A total of 46 abnormal events were identified and categorized according to their deviation characteristics. These events consisted of both unusually high consumption occurrences and periods of unexpectedly low electricity usage. For each event, information such as the occurrence date, deviation score, event category, and associated appliance was recorded and displayed for further examination.

The adopted Z-score methodology enables continuous monitoring of consumption behaviour by comparing current observations with recent historical trends. Positive deviation values correspond to higher-than-expected electricity usage, whereas negative values indicate a reduction relative to normal operating conditions. Events exceeding the predefined threshold limits are automatically flagged for attention.

The detected abnormalities were most frequently associated with appliances such as the Washing Machine and Refrigerator. These devices often exhibit varying operating cycles and changing usage frequencies, which can occasionally produce significant departures from typical consumption behaviour. The generated visualizations assist in identifying such events and facilitate further investigation of their underlying causes.

The anomaly analysis results demonstrate the ability of the developed framework to recognize unusual usage patterns within household electricity data. Such information can support early identification of inefficient operation, unexpected consumption behaviour, or potential equipment-related concerns, thereby contributing to more effective electricity usage monitoring and cost management.

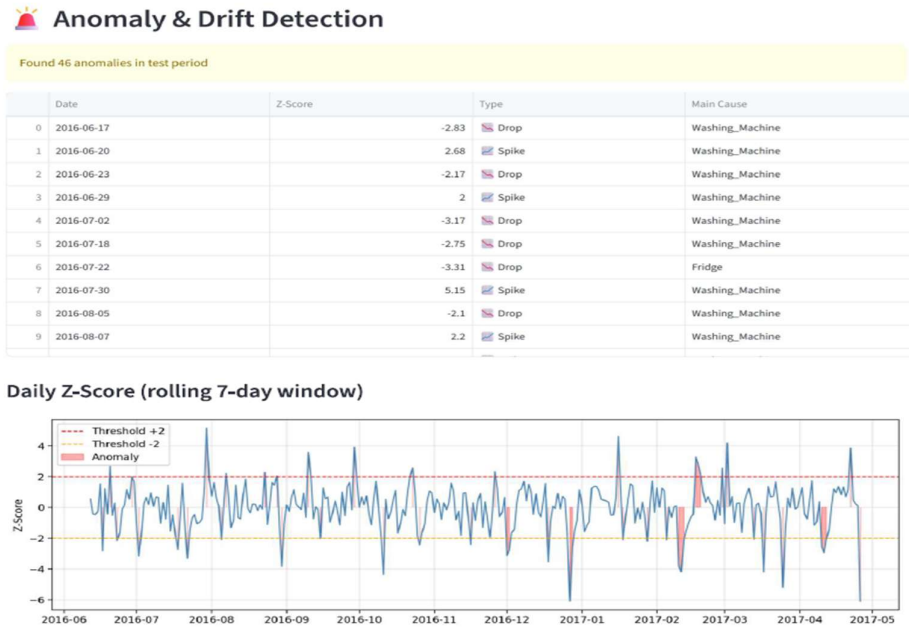


Figure 13 : Anomaly and Drift Detection

6.7 Electricity Bill Estimation Results

An expenditure analysis was performed to estimate the monthly electricity cost associated with the predicted household consumption profile. The calculation process utilizes projected monthly energy usage together with user-defined tariff rates, enabling cost estimation under different pricing structures.

Figures 14 and 15 present the expenditure analysis results for both United Kingdom and Indian tariff settings. The visualization interface allows tariff values to be adjusted dynamically, after which the corresponding monthly cost is recalculated automatically.

For the selected evaluation period, the projected monthly electricity usage was approximately 908.72 kWh. Based on a tariff rate of 28 pence per kWh, the estimated monthly expenditure was calculated as £254.44. When an Indian tariff rate of ₹7 per kWh was applied, the corresponding monthly cost was estimated at ₹6361.07.

A detailed appliance-wise cost distribution was also generated to identify the major contributors to overall expenditure. The analysis indicates that the Refrigerator and Kettle account for the largest proportion of monthly electricity costs, whereas the Dishwasher contributes the smallest share among the evaluated appliances. Such information can help users recognize the devices responsible for a significant portion of household electricity expenses.

The generated cost estimates provide a practical understanding of how electricity consumption translates into financial expenditure under different tariff conditions. By linking consumption patterns with monetary impact, the analysis supports greater awareness of household electricity usage and assists users in identifying opportunities for cost reduction.

Overall, the results demonstrate the capability of the developed framework to provide meaningful expenditure estimates and appliance-wise cost insights using predicted consumption information.

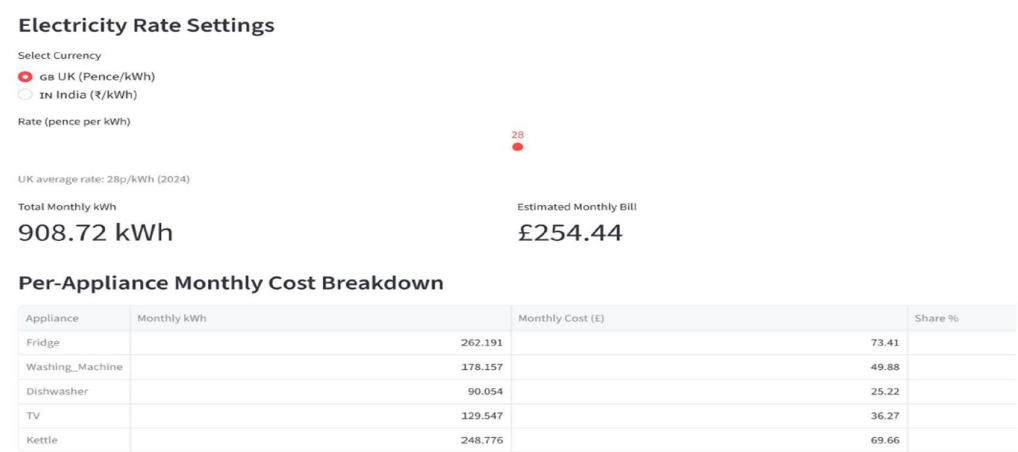


Figure 14 : Electricity Bill Estimation in GB UK

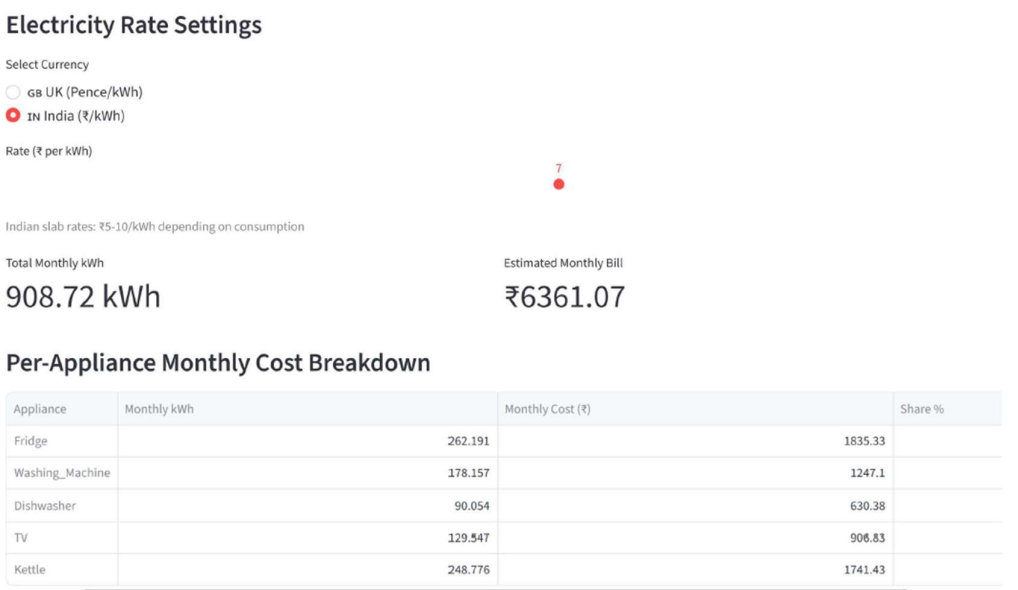


Figure 15 : Electricity Bill Estimation in Indian Rupee

6.8 Discussion

The experimental findings indicate that the developed learning framework is capable of extracting meaningful device-level information from combined household electricity measurements. The obtained results demonstrate that prediction quality is strongly influenced by the operating characteristics of individual appliances and the distinctiveness of their consumption behaviour.

Devices exhibiting short-duration and clearly identifiable usage patterns generally achieved superior performance. Such appliances generate electricity demand profiles that are easier to distinguish from the overall household load, enabling more accurate recognition of operating states and consumption events. Conversely, appliances characterized by cyclic operation, variable usage duration, or overlapping demand patterns produced comparatively lower classification performance. These observations are consistent with challenges reported in previous energy disaggregation studies and highlight the inherent difficulty of separating similar consumption signatures from a single measurement source [1], [2], [4].

The evaluation results further indicate that satisfactory estimation accuracy can be achieved even when state-recognition performance varies among appliances. This suggests that the framework is capable of capturing broader consumption trends and operational behaviour despite the complexity associated with certain device categories.

Beyond consumption estimation, the additional analytical components enhanced the practical usefulness of the developed framework. The anomaly analysis functionality successfully identified unusual usage events, providing information that may assist in detecting unexpected consumption behaviour or inefficient appliance operation. Similarly, the expenditure analysis feature translated electricity usage into monetary terms, allowing users to better understand the financial impact of household consumption patterns.

The visualization environment contributed significantly to the interpretability of the generated results. By presenting statistical summaries, consumption trends, expenditure information, and analytical outputs within a unified interface, the system enables easier exploration of electricity usage behaviour and supports data-driven decision-making.

Overall, the findings demonstrate that the developed framework provides a practical approach for residential electricity analysis using a single source of household consumption data. The combination of consumption estimation, abnormal usage identification, expenditure assessment, and interactive visualization extends the functionality beyond conventional prediction tasks and improves the overall usefulness of the developed solution.

Chapter 7

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This work presented the development of a residential electricity analytics framework capable of estimating device-level consumption information from combined household electricity measurements. By utilizing deep learning techniques and Non-Intrusive Load Monitoring principles, the developed approach demonstrated the ability to extract meaningful insights without requiring dedicated monitoring hardware for individual appliances.

The experimental evaluation showed that the framework successfully identified consumption characteristics across multiple household devices and achieved particularly strong performance for appliances with distinctive operating patterns. The obtained results confirmed the effectiveness of the adopted learning approach in recognizing usage behaviour and generating reliable consumption estimates from household electricity records.

In addition to consumption estimation, the developed framework incorporated analytical capabilities such as abnormal usage identification and expenditure assessment. These features enhanced the practical value of the system by providing users with information related to unusual consumption events and the financial impact of electricity usage.

A web-based visualization environment was also developed to present analytical outputs through interactive charts, summaries, and monitoring tools. The integration of prediction, analysis, and visualization components enabled the creation of a unified platform for examining household electricity behaviour.

Overall, the objectives of the study were successfully achieved. The developed framework demonstrates the potential of data-driven approaches for residential electricity analysis and provides a foundation for supporting energy awareness, consumption optimization, and informed decision-making in modern households.

7.2 Future Scope

While the developed framework produced encouraging results, several opportunities exist for further improvement and expansion. Future research can focus on enhancing prediction performance, increasing deployment flexibility, and extending the range of analytical capabilities offered by the system.

Potential directions for future work include:

- Integration with smart meters and Internet of Things (IoT) devices to support continuous real-time monitoring.
- Deployment on edge-computing platforms such as Raspberry Pi and embedded hardware for low-latency operation.
- Investigation of advanced neural network architectures, including Transformer and attention-based approaches, for improved prediction accuracy.
- Expansion of the framework to accommodate a larger variety of household appliances and consumption scenarios.
- Development of personalized recommendations that assist users in reducing electricity usage and improving efficiency.
- Incorporation of predictive maintenance features for early identification of abnormal appliance behaviour and potential equipment failures.
- Implementation of cloud-based services to enable remote access, centralized data management, and multi-user support.
- Enhancement of anomaly analysis through more sophisticated statistical and machine learning techniques.
- Integration of renewable energy components, including solar generation and battery storage monitoring.
- Evaluation using region-specific datasets and tariff structures to improve adaptability across different geographical locations.

The incorporation of these enhancements can increase the practicality, scalability, and intelligence of future residential electricity analysis platforms. Such developments may contribute to more efficient energy utilization and support the evolution of smart homes and next-generation energy management systems.

GANTT CHART

Activity	Time Frame (in Months)				
	January 2026	February 2026	March 2026	April 2026	May 2026
Literature Survey					
Problem Definition					
Implementation					
Testing And Validation					
Report writing					

Activity Proposed



Activity Achieved



Figure 16 : Timeline and Milestones for AI-Based Smart Energy Monitoring System

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